

Self-Check Questions: Integration

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1 Riemann and Darboux Integration

1.1 Foundational Definitions

1. What is a partition of an interval $[a, b]$?
2. Define the norm (mesh size) of a partition.
3. What is a tagged partition?
4. Define a Riemann sum.
5. State the definition of the Riemann integral using Riemann sums.
6. Define the upper Darboux sum.
7. Define the lower Darboux sum.
8. What are the quantities

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

and

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)?$$

9. Define the upper Darboux integral.
10. Define the lower Darboux integral.
11. State the definition of Darboux integrability.
12. Why must a function be bounded to be Riemann integrable?
13. Why must a function be bounded to be Darboux integrable?

1.2 Conceptual Understanding

14. What is the geometric intuition behind a Riemann sum?
15. What is the geometric intuition behind upper and lower Darboux sums?
16. Why does refining a partition generally improve approximation quality?
17. What happens to lower Darboux sums when a partition is refined?
18. What happens to upper Darboux sums when a partition is refined?
19. Why is

$$L(f, P) \leq U(f, P)$$

always true?

20. Explain why

$$\sup_P L(f, P) \leq \inf_P U(f, P).$$

21. What does it mean geometrically when the upper and lower integrals are equal?
22. Why are Darboux sums often considered technically simpler than Riemann sums?
23. Why are Riemann sums closer to the historical intuition of integration?

1.3 Comparisons Between Riemann and Darboux Integration

24. What is the main difference between a Riemann sum and a Darboux sum?
25. In a Riemann sum, how are sample points chosen?
26. In a Darboux sum, why are sample points unnecessary?
27. Which theory uses arbitrary sample points explicitly?
28. Which theory uses suprema and infima on subintervals?
29. What role does the partition norm play in Riemann integration?
30. What role does partition refinement play in Darboux integration?
31. State the theorem relating Riemann integrability and Darboux integrability.
32. Why are the two theories equivalent for bounded functions on closed intervals?
33. Can two different tagged partitions with the same underlying partition produce different Riemann sums?
34. Can two different Darboux sums exist for the same partition?
35. How do upper and lower Darboux sums bound all possible Riemann sums?
36. Explain how Darboux integration avoids explicit limits of sums.
37. Explain how Riemann integration encodes convergence directly through limits.

1.4 Theorems and Properties

38. State the linearity property of the integral.
39. State the additivity over intervals property.
40. State the monotonicity property of the integral.
41. If f is continuous on $[a, b]$, why is it integrable?
42. Why is every monotone function on $[a, b]$ integrable?
43. State a criterion for integrability involving discontinuities.
44. What is the relationship between measure-zero discontinuity sets and integrability?
45. Why is the Dirichlet function not Riemann integrable?
46. Why is the characteristic function of the rationals discontinuous everywhere?
47. Give an example of a function that is discontinuous but still Riemann integrable.
48. Why is a function with finitely many discontinuities integrable?
49. What happens to the integral if two functions differ only on finitely many points?
50. What happens if two functions differ only on a measure-zero set?

1.5 Technical Manipulation

51. Compute the lower and upper Darboux sums for a constant function.
52. Compute the lower and upper Darboux sums for $f(x) = x$ on $[0, 1]$ using a uniform partition.

53. Derive the Riemann sum for $f(x) = x^2$ on $[0, 1]$.

54. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Riemann sums.

55. Show directly that

$$\int_0^1 x \, dx = \frac{1}{2}$$

using Darboux sums.

56. For a uniform partition, how do you compute

$$\Delta x?$$

57. Why do continuous functions attain maxima and minima on closed intervals?

58. Why is compactness important in Darboux integration?

59. What is the difference between taking a supremum over all lower sums and taking a limit?

60. Why does the existence of arbitrarily close upper and lower sums imply integrability?

1.6 Interpreting the Integral

61. What does the expression

$$\sum_i f(x_i) \Delta x_i$$

represent before assigning any geometric or physical interpretation to it?

62. Why is the integral fundamentally an accumulation operator rather than specifically an area operator?

63. In the area interpretation of the integral, what do the quantities

$$f(x_i), \quad \Delta x_i, \quad f(x_i) \Delta x_i$$

represent geometrically?

64. Why does

$$f(x_i) \Delta x_i$$

approximate the area of a thin rectangle?

65. What happens to the Riemann sum as

$$\max \Delta x_i \rightarrow 0?$$

66. Why does the definite integral emerge as a limit of Riemann sums?

67. If

$$g'(x) = f(x),$$

why does

$$f(x) \Delta x$$

approximate the small change

$$\Delta g?$$

68. What is the conceptual meaning of

$$dg = g'(x) dx?$$

69. Why can the same expression

$$f(x)\Delta x$$

be interpreted both as rectangle area and as accumulated change?

70. What determines the interpretation of the integrand $f(x)$?

71. Why is the equality

$$\int_a^b g'(x) dx = g(b) - g(a)$$

an accumulation statement?

72. In the Fundamental Theorem of Calculus, what exactly is being accumulated?

73. Why does integrating a derivative recover net change?

74. How does the telescoping intuition appear in

$$\sum \Delta g_i?$$

75. Define

$$A(x) = \int_a^x f(t) dt.$$

What does $A(x)$ represent geometrically?

76. Why does increasing x by a small amount h add approximately

$$f(x)h$$

to the accumulated area?

77. Use the difference quotient to explain why

$$A'(x) = f(x).$$

78. Why does the identity

$$A'(x) = f(x)$$

connect the area interpretation to the antiderivative interpretation?

79. Why is the Fundamental Theorem of Calculus fundamentally a statement that differentiation and integration are inverse operations?

80. How does the notion of “local rate” connect to “global accumulation”?

81. What is the difference between:

- a function value,
- a derivative,
- a differential,
- an accumulated quantity?

82. Why is

$$\int_a^b f(x) dx$$

not intrinsically an area?

83. Give examples where the integral represents:

- distance,
- mass,
- charge,
- probability,
- energy.

84. In each example above, what plays the role of:

- density/rate,
- small measure,
- accumulated total?

85. Explain the pattern:

$$\text{density} \times \text{small measure} \rightarrow \text{accumulated quantity}.$$

86. Why is the rectangle picture only one geometric realization of integration?

87. What is the conceptual difference between:

$$f(x) \quad \text{and} \quad f(x) dx?$$

88. Why is

$$dx$$

best understood as “a tiny increment of the independent variable”?

89. Why is the statement

$$dg = g'(x) dx$$

more fundamental than the rectangle interpretation?

90. What does the Fundamental Theorem of Calculus reveal about the relationship between local structure and global structure?

2 Cauchy Principal Value

1. What problem is the Cauchy principal value trying to solve?
2. Why does

$$\int_{-1}^1 \frac{1}{x} dx$$

fail to exist as an ordinary improper integral?

3. What is the difference between an improper integral and a principal value?
4. Why is a principal value not simply another way to evaluate an improper integral?
5. What does the word “symmetric” mean in the definition of principal value?
6. Why is symmetry essential in the definition?
7. Can a principal value exist when the corresponding improper integral does not?
8. Can an improper integral exist when the principal value also exists?
9. What information is lost when we replace an improper integral by a principal value?
10. What geometric intuition can be attached to principal values?

2.1 Computational Questions

11. Compute

$$\text{PV} \int_{-1}^1 \frac{1}{x} dx.$$

12. Why does the answer not contradict the fact that the ordinary improper integral diverges?
13. Compute

$$\text{PV} \int_{-2}^3 \frac{1}{x} dx.$$

14. Why is the answer no longer zero?
15. Derive a general formula for

$$\text{PV} \int_a^b \frac{1}{x} dx,$$

assuming $a < 0 < b$.

16. Compute

$$\text{PV} \int_{-1}^1 \frac{x}{x^2 + 1} dx.$$

17. Compute

$$\text{PV} \int_{-1}^1 \frac{1+x}{x} dx.$$

18. Why is it useful to split the integrand into a singular part and a regular part?
19. Compute

$$\text{PV} \int_{-1}^1 \frac{\sin x}{x} dx.$$

20. Is a principal value always finite when the singularity is of the form $1/x$?

2.2 Questions About Singularities

21. Why does $1/x$ behave differently from $1/x^2$?
22. Does

$$\text{PV} \int_{-1}^1 \frac{1}{x^2} dx$$

exist?

23. If not, why not?
24. What property of $1/x$ allows cancellation?
25. What property of $1/x^2$ prevents cancellation?
26. For which powers p does

$$\text{PV} \int_{-1}^1 \frac{1}{x^p} dx$$

exist?

27. What role does oddness or evenness of a function play?
28. When does symmetry force a principal value to be zero?
29. Can an odd function have a divergent principal value?
30. Can an even function have a finite principal value?

2.3 Questions About Infinity

31. What is

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x} dx?$$

32. How does this differ from

$$\int_{-\infty}^{\infty} \frac{1}{x} dx?$$

33. Why must limits at $+\infty$ and $-\infty$ be taken symmetrically?
34. What happens if the two limits are taken independently?
35. Give an example where symmetric truncation gives a finite value but ordinary convergence fails.
36. Why is

$$\text{PV} \int_{-\infty}^{\infty} \sin x dx$$

an interesting example?

37. How does principal value at infinity differ from principal value around an interior singularity?

2.4 Theory Questions

38. State the formal definition of the Cauchy principal value for an interior singularity.
39. State the formal definition for an integral over the whole real line.
40. Why is the principal value defined as a limit?

41. Under what conditions does the principal value agree with the ordinary improper integral?
42. Is the principal value linear?
43. When is it legitimate to split a principal value integral into two pieces?
44. Can principal values always be manipulated algebraically as ordinary integrals?
45. What pitfalls arise when changing variables in principal value integrals?
46. Why must one be careful about symmetry when performing substitutions?
47. Is principal value integration compatible with all standard integration theorems?
48. Which familiar theorems from elementary calculus continue to work unchanged?
49. Which require extra hypotheses?
50. Why do analysts treat principal values as generalized notions of integration?

3 Integrating Inverse Functions

1. Let $f : I \rightarrow J$ be one-to-one. What does it mean for $f^{-1} : J \rightarrow I$ to be the inverse of f ? State both identities involving f and f^{-1} .
2. Why is injectivity of f necessary if we want f^{-1} to be an ordinary function on $J = f(I)$?
3. Suppose $x = f(t)$. Explain why

$$f^{-1}(x) = t.$$

4. In the informal substitution

$$\int f^{-1}(x) dx = \int t f'(t) dt, \quad x = f(t),$$

which two transformations are being used?

5. Why should the expression

$$dx = f'(t) dt$$

not be regarded as the fundamental rigorous justification of the substitution?

6. What theorem provides the rigorous justification for the transformation

$$\int f^{-1}(x) dx \longrightarrow \int f^{-1}(f(t)) f'(t) dt?$$

7. Let F be an antiderivative of f^{-1} . Write explicitly what the statement

$$F' = f^{-1}$$

means.

8. Differentiate $F(f(t))$ using the chain rule.

9. Starting from

$$\frac{d}{dt} F(f(t)),$$

derive in a few lines the identity

$$\frac{d}{dt} F(f(t)) = t f'(t).$$

10. Why does the identity

$$(F \circ f)'(t) = t f'(t)$$

show that $F \circ f$ is an antiderivative of $t f'(t)$?

11. Explain carefully what is meant by the informal identity

$$\int f^{-1}(x) dx = \int t f'(t) dt.$$

In particular, why are the two antiderivatives being written in different variables?

12. State a precise definite-integral version of

$$\int f^{-1}(x) dx = \int t f'(t) dt.$$

13. Prove that for $a, b \in I$,

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = \int_a^b t f'(t) dt.$$

14. Derive the same definite-integral identity directly from the substitution theorem by setting

$$x = f(t).$$

15. In the substitution

$$x = f(t),$$

what happens to the lower and upper bounds $f(a)$ and $f(b)$?

16. Does the identity

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = \int_a^b t f'(t) dt$$

require f to be increasing? Explain.

17. Suppose f is decreasing. What is the relative order of $f(a)$ and $f(b)$ when $a < b$, and why does the definite-integral formula still remain valid?

18. Apply integration by parts to

$$\int_a^b t f'(t) dt.$$

State clearly your choices of u and dv .

19. Derive

$$\int_a^b t f'(t) dt = b f(b) - a f(a) - \int_a^b f(t) dt.$$

20. Combine the inverse-function substitution formula with integration by parts to prove

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = b f(b) - a f(a) - \int_a^b f(t) dt.$$

21. Let $x \in J$ and put

$$b = f^{-1}(x).$$

Show that $f(b) = x$.

22. Starting from

$$\int_{f(a)}^{f(b)} f^{-1}(u) du = b f(b) - a f(a) - \int_a^b f(t) dt,$$

substitute $b = f^{-1}(x)$ and derive

$$\int_{f(a)}^x f^{-1}(u) du = x f^{-1}(x) - a f(a) - \int_a^{f^{-1}(x)} f(t) dt.$$

23. Why is the previous formula more precise than the informal identity

$$\int f^{-1}(x) dx = x f^{-1}(x) - \int f(t) dt?$$

24. In the informal formula

$$\int f^{-1}(x) dx = x f^{-1}(x) - \int f(t) dt,$$

what is ambiguous about the last integral?

25. Give a corrected indefinite-integral formula in which the dependence on x is explicit.

26. Let

$$G(x) = xf^{-1}(x) - \int_a^{f^{-1}(x)} f(t) dt.$$

Differentiate $G(x)$ directly and verify that

$$G'(x) = f^{-1}(x).$$

27. In the previous calculation, which rule is needed to differentiate

$$\int_a^{f^{-1}(x)} f(t) dt?$$

28. Assuming $(f^{-1})'(x)$ exists, carry out the derivative of

$$xf^{-1}(x)$$

using the product rule.

29. Show explicitly how the terms involving $(f^{-1})'(x)$ cancel when differentiating

$$xf^{-1}(x) - \int_a^{f^{-1}(x)} f(t) dt.$$

30. Why does the direct differentiation proof of the inverse-function integration formula typically require stronger assumptions than the proof based on

$$(F \circ f)'(t) = tf'(t)?$$

31. What hypotheses are sufficient to ensure that the substitution theorem can be applied to

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx$$

with $x = f(t)$?

32. Which assumptions are needed merely to define f^{-1} , and which additional assumptions are needed for the differentiability or integration arguments?

33. Suppose $f(t) = t^2$ on $[0, \infty)$. Find $f^{-1}(x)$ and verify

$$\int f^{-1}(x) dx = \int tf'(t) dt$$

by direct computation.

34. For $f(t) = t^2$ on $[0, \infty)$, derive the usual formula for

$$\int \sqrt{x} dx$$

using the inverse-function method.

35. Suppose $f(t) = e^t$. Use the formula

$$\int f^{-1}(x) dx = \int tf'(t) dt$$

to derive an antiderivative of $\ln x$.

36. Starting from $f(t) = e^t$, use integration by parts to derive

$$\int \ln x \, dx = x \ln x - x + C.$$

37. Suppose $f(t) = \tan t$ on $(-\pi/2, \pi/2)$. Use the inverse-function method to express

$$\int \arctan x \, dx$$

as an integral in the variable t .

38. Complete the derivation of

$$\int \arctan x \, dx$$

from the previous question.

39. Suppose $f(t) = \sin t$ on $[-\pi/2, \pi/2]$. Use the inverse-function method to derive a formula for

$$\int \arcsin x \, dx.$$

40. Why must the domain of $\sin t$ be restricted before treating $\arcsin$ as its inverse?

41. Suppose $f(t) = t^3 + t$. Without finding an explicit formula for f^{-1} , express

$$\int_{f(a)}^{f(b)} f^{-1}(x) \, dx$$

in terms of a , b , and an elementary integral involving f .

42. For $f(t) = t^3 + t$, compute

$$\int_{f(0)}^{f(1)} f^{-1}(x) \, dx$$

without explicitly solving for f^{-1} .

43. Explain geometrically why an identity involving

$$\int f(x) \, dx \quad \text{and} \quad \int f^{-1}(x) \, dx$$

should be plausible when f is increasing.

44. If the graph of $y = f(x)$ is reflected across the line $y = x$, what graph is obtained?

45. For an increasing positive function f , explain how the rectangle with side lengths b and $f(b)$ can be decomposed into areas associated with f and f^{-1} .

46. Derive the geometric area identity

$$\int_a^b f(t) \, dt + \int_{f(a)}^{f(b)} f^{-1}(x) \, dx = bf(b) - af(a).$$

47. Which part of the rigorous analytic proof corresponds geometrically to the fact that the graphs of f and f^{-1} are reflections of each other?

48. Explain the role of the fundamental theorem of calculus in passing from an antiderivative statement to a definite-integral identity.

49. Explain the role of the chain rule in the proof.
50. Explain the role of integration by parts in obtaining a formula involving both f and f^{-1} .
51. Distinguish clearly between the following three statements:

$$(F \circ f)'(t) = tf'(t),$$

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = \int_a^b tf'(t) dt,$$

and

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a) - \int_a^b f(t) dt.$$

What theorem or technique is responsible for each transition?

52. Give a complete proof, from the chain rule and the fundamental theorem of calculus, of

$$\int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a) - \int_a^b f(t) dt.$$

53. Reconstruct the entire argument without using the symbolic differential equation

$$dx = f'(t) dt.$$

54. Finally, explain in your own words why the informal substitution

$$x = f(t), \quad dx = f'(t) dt$$

works in this problem, and identify the rigorous theorem that justifies each symbolic step.

4 Differentiation Under the Integral Sign

1. What is the statement of Leibniz's rule for differentiating under the integral sign with constant limits?
2. What is the general form of Leibniz's rule when the limits of integration depend on the parameter?
3. What conditions must be satisfied to justify differentiation under the integral sign (continuity, uniform convergence, dominated convergence, etc.)?
4. What role does continuity of the integrand and its partial derivative play in applying differentiation under the integral sign?
5. How does the Dominated Convergence Theorem justify differentiation under the integral sign?
6. When can uniform convergence be used to justify differentiation under the integral sign?
7. What is the difference between pointwise convergence and uniform convergence in this context?
8. How do you compute $\frac{d}{d\alpha} \int_a^b f(x, \alpha), dx$?
9. How do you compute $\frac{d}{d\alpha} \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha), dx$?
10. What additional terms appear when the limits of integration depend on the parameter?
11. How do you apply differentiation under the integral sign to evaluate difficult integrals?
12. What is the idea behind Feynman's technique for evaluating integrals using parameters?
13. How can introducing a parameter simplify the evaluation of an integral?
14. Give an example where differentiating under the integral sign transforms an integral into a simpler one.
15. How do you recover the original integral after differentiating with respect to a parameter?
16. What initial or boundary conditions are needed after integrating back with respect to the parameter?
17. How do you handle constants of integration when using this method?
18. When is it valid to interchange differentiation and integration?
19. What are common pitfalls when applying differentiation under the integral sign?
20. How does this technique apply to improper integrals?
21. What extra care is needed when the interval of integration is infinite?
22. How do you justify differentiation under the integral sign for $\int_0^\infty f(x, \alpha), dx$?
23. Can differentiation under the integral sign be applied multiple times? Under what conditions?
24. What is the relationship between differentiation under the integral sign and parameter-dependent integrals?
25. How does this method connect to solving differential equations?
26. How can this technique be used in probability theory (e.g., moment generating functions)?

27. How does differentiation under the integral sign relate to Fourier transforms or Laplace transforms?
28. What are some classic integrals that are evaluated using this method?
29. How would you construct your own parameterized integral to apply this technique?
30. How do you verify that your final answer is correct after applying the method?

5 Cauchy Formula for Repeated Integration

5.1 Basic understanding

1. What problem does the Cauchy formula for repeated integration solve?
2. What is meant by the " n -th repeated integral" of a function?
3. Write the definition of the n -fold repeated integral of a function f with the base point a :

$$f^{(-n)}(x) = ?$$

4. Under what conditions on f does the formula hold?
5. What is the main result of the Cauchy formula for repeated integration? State it explicitly.

5.2 Formula and structure

6. Rewrite the repeated integral as a single integral:

$$f^{(-n)}(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt.$$

7. What is the role of the factor $\frac{1}{(n-1)!}$?
8. Why does the kernel $(x-t)^{n-1}$ appear in the formula?
9. What happens in the special case $n = 1$? Verify that the formula reduces correctly.
10. Compute explicitly $f^{(-2)}(x)$ using both:
 - the definition (nested integrals),
 - the Cauchy formula,and verify that they agree.

5.3 Proof understanding

11. What proof technique is used to prove the formula?
12. What is the base case of the induction?
13. Show that for $n = 1$:

$$f^{(-1)}(x) = \int_a^x f(t) dt.$$

14. What assumption is made in the induction step?
15. What must be proven in the induction step?
16. Where is the Leibniz integral rule used in the proof?
17. Compute:

$$\frac{d}{dx} \left[\frac{1}{n!} \int_a^x (x-t)^n f(t) dt \right]$$

and explain why it gives the desired recursive relation.

18. How does the induction step transform an $(n+1)$ -fold integral into the desired single-integral form?
19. Why does the induction argument prove the formula for all $n \in \mathbb{N}$?

5.4 Conceptual understanding

20. Why can repeated integration be "compressed" into a single integral?
21. How is this formula related to convolution?
22. Interpret the kernel $(x - t)^{n-1}$ geometrically or intuitively.
23. How does this formula simplify practical computations?
24. In what situations is this formula especially useful?

5.5 Generalizations

25. How can the formula be extended to non-integer orders of integration?
26. What replaces the factorial $(n - 1)!$ in the generalized version?
27. State the fractional integral version:

$$(J^\alpha f)(x) = ?$$

28. What condition must α satisfy?
29. What is the name of this generalization?
30. How does the formula relate to fractional derivatives (when $\alpha < 0$)?

5.6 Connections and further ideas

31. How is this formula connected to fractional calculus?
32. What is a differintegral?
33. How can repeated integration be used to define differentiation of non-integer order?
34. How is this formula generalized to higher dimensions?
35. What is the Riesz potential?

5.7 Practice problems

36. Compute:

$$\int_0^x \int_0^{t_1} t_2 \, dt_2 \, dt_1$$

using the Cauchy formula.

37. Compute the n -fold integral of $f(t) = 1$.
38. Compute the n -fold integral of $f(t) = t^k$.
39. Show that:

$$(J^n f)(x) = (f * g_n)(x)$$

for an appropriate kernel g_n .

40. Verify the formula numerically for a simple function (e.g. $f(t) = e^t$).

6 Frullani's Integral

1. State Frullani's integral theorem.
2. What conditions on $f(x)$ are typically required for Frullani's theorem to hold?
3. Why must the limits $f(0)$ and $f(\infty)$ exist?
4. What is the role of the condition $a, b > 0$?
5. Why does the logarithm naturally appear in Frullani-type integrals?
6. Derive

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \ln \frac{b}{a}.$$

7. Explain why

$$\int_0^\infty \frac{e^{-ax}}{x} dx$$

diverges.

8. Why does subtracting two individually divergent integrals sometimes produce a convergent result?
9. What happens if $a = b$?
10. What happens if one of the parameters tends to zero?
11. What happens if one of the parameters tends to infinity?
12. Show how Frullani's theorem can be derived using differentiation under the integral sign.
13. Define

$$F(a) = \int_0^\infty \frac{e^{-x} - e^{-ax}}{x} dx.$$

Compute $F'(a)$, use the initial condition $F(1) = 0$ to determine the constant of integration, and write down the closed form for $F(a)$.

14. When applying the Feynman trick to

$$I(a) = \int_0^\infty \frac{f(x) - f(ax)}{x} dx,$$

what initial condition pins down the constant of integration? Why does this condition hold?

15. Differentiate

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$$

with respect to a , solve the resulting equation, and verify that the answer matches Frullani's theorem.

16. In the Feynman approach, differentiating with respect to a removes the $\frac{1}{x}$ factor from the integrand. Why does this make the integral tractable, and what condition on f justifies differentiating under the integral sign?
17. Why does the final answer depend only on $f(0)$ and $f(\infty)$ rather than the full shape of f ?
18. Give an example of a function for which Frullani's theorem fails, and identify which hypothesis is violated.

19. Derive Frullani's theorem from the identity

$$f(ax) - f(bx) = \int_b^a x f'(tx) dt.$$

20. Compute

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx.$$

21. Compute

$$\int_0^\infty \frac{\ln(1+ax) - \ln(1+bx)}{x} dx.$$

Verify the answer by checking the $a = b$ limit.

22. Does Frullani's theorem apply to $\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx$? Identify which hypothesis is (or is not) satisfied, and compute the integral by another method.

23. What changes if the integration interval is $(0, T)$ rather than $(0, \infty)$?

24. Write $\frac{1}{x} = \int_0^\infty e^{-xs} ds$ and substitute into $\int_0^\infty \frac{f(ax) - f(bx)}{x} dx$. After a formal interchange of the order of integration, express the result in terms of the Laplace transform $\mathcal{L}\{f\}(s) = \int_0^\infty f(x)e^{-sx} dx$.

25. Show that

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \int_0^\infty \left[\frac{1}{a} \mathcal{L}\{f\}\left(\frac{s}{a}\right) - \frac{1}{b} \mathcal{L}\{f\}\left(\frac{s}{b}\right) \right] ds.$$

Verify this for $f(x) = e^{-x}$ by computing the right-hand side directly and confirming the answer equals $\ln(b/a)$.

26. State the identity

$$\int_0^\infty \frac{g(x)}{x} dx = \int_0^\infty \mathcal{L}\{g\}(s) ds$$

and give sufficient conditions for its validity. Use it to rederive Frullani's theorem in one step by setting $g(x) = f(ax) - f(bx)$ and computing $\mathcal{L}\{g\}(s)$ explicitly.

27. In the Laplace-transform approach, why is the condition $g(0) = 0$ necessary for the identity in the previous question? Show that $g(0) = 0$ follows automatically when $g(x) = f(ax) - f(bx)$ and the hypotheses of Frullani's theorem are satisfied.
28. What justifies the interchange of the order of integration in the Laplace-transform proof? State the theorem you invoke (e.g. Tonelli's theorem) and verify its hypotheses for $f(x) = e^{-x}$.
29. How is Frullani's theorem connected to Laplace transforms in a broader sense? (*Hint*: consider the formula $\int_0^\infty f(x)/x dx = \int_0^\infty F(s) ds$, where $F = \mathcal{L}\{f\}$, and how it applies once the condition $g(0) = 0$ is ensured.)
30. Both the Feynman trick and the Laplace-kernel method deal with the $\frac{1}{x}$ factor. In the Feynman approach, $\frac{1}{x}$ is absorbed by differentiating with respect to a parameter; in the Laplace approach, $\frac{1}{x}$ is represented as $\int_0^\infty e^{-xs} ds$. Explain the structural analogy between these two approaches, and describe a setting in which one is preferable to the other.
31. What assumptions are required to differentiate under the integral sign? State a sufficient condition (e.g. dominated convergence) and verify it for $f(x) = e^{-x}$ in the Feynman proof of Frullani's theorem.

7 Dirichlet's Integral

1. State Dirichlet's integral:

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What is its value?

2. Why is the integrand not actually singular at $x = 0$? Compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

3. Does

$$\int_0^\infty \left| \frac{\sin x}{x} \right| dx$$

converge? What does this tell you about the convergence of Dirichlet's integral?

4. Use integration by parts on

$$\int_1^R \frac{\sin x}{x} dx$$

to show that the limit as $R \rightarrow \infty$ exists.

5. For $a > 0$, compute

$$\int_0^\infty \frac{\sin(ax)}{x} dx$$

by the substitution $u = ax$.

6. What happens to the previous integral when $a = 0$?

7. What happens when $a < 0$? Use the oddness of $\sin$ to determine the sign.

8. Compare

$$\int_0^\infty \frac{\sin(ax)}{x} dx \quad \text{and} \quad \int_0^\infty \sin(ax) dx.$$

Why does the factor $\frac{1}{x}$ matter so much?

9. Define

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx, \quad a > 0.$$

Why is this integral easier to handle than

$$\int_0^\infty \frac{\sin x}{x} dx$$

directly?

10. Differentiate $F(a)$ with respect to a and show that

$$F'(a) = - \int_0^\infty e^{-ax} \sin x dx.$$

11. Compute

$$\int_0^\infty e^{-ax} \sin x dx$$

for $a > 0$.

12. Deduce that

$$F'(a) = -\frac{1}{1+a^2}.$$

Integrate this with respect to a .

13. Use the condition

$$\lim_{a \rightarrow \infty} F(a) = 0$$

to determine the constant of integration.

14. Show that

$$F(a) = \frac{\pi}{2} - \arctan a = \arctan \frac{1}{a}.$$

15. Let $a \rightarrow 0^+$ and recover Dirichlet's integral.

16. Why is introducing the factor e^{-ax} a useful trick before letting $a \rightarrow 0^+$?

17. Verify the identity

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt.$$

18. Substitute this identity into

$$\int_0^\infty \frac{\sin x}{x} dx.$$

What formal double integral do you obtain?

19. Why is it dangerous to interchange the order of integration in that double integral without extra justification?

20. Explain why

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

can be viewed as the Laplace transform of $\frac{\sin x}{x}$ evaluated at a .

21. If

$$f(x) = \frac{\sin x}{x},$$

write $F(a) = \mathcal{L}\{f\}(a)$. What does the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

tell you about $\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$?

22. Starting from

$$\frac{\sin x}{x} = \int_0^1 \cos(tx) dt,$$

use linearity of the Laplace transform to derive

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \mathcal{L}\{\cos(tx)\}(a) dt.$$

Why is this a safer way to use the identity than working directly with

$$\int_0^\infty \frac{\sin x}{x} dx?$$

23. Compute

$$\mathcal{L}\{\cos(tx)\}(a)$$

for $a > 0$, and use it to show that

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \int_0^1 \frac{a}{a^2 + t^2} dt.$$

In other words, show that

$$F(a) = \int_0^1 \int_0^\infty e^{-ax} \cos(tx) dx dt.$$

24. Evaluate

$$\int_0^1 \frac{a}{a^2 + t^2} dt$$

and recover the formula

$$\mathcal{L}\left\{\frac{\sin x}{x}\right\}(a) = \frac{\pi}{2} - \arctan a.$$

25. Use the Laplace-transform identity

$$\mathcal{L}\{f'\}(a) = a\mathcal{L}\{f\}(a) - f(0)$$

with

$$f(x) = \frac{\sin x}{x}$$

to explain informally why the transform of $\frac{\sin x}{x}$ should be related to an arctangent.

26. Why does the limit

$$\lim_{a \rightarrow 0^+} \mathcal{L}\left\{\frac{\sin x}{x}\right\}(a)$$

recover Dirichlet's integral?

27. Compare these two viewpoints: differentiating

$$F(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$$

with respect to a , and recognizing $F(a)$ as a Laplace transform from the start. Why are these really two versions of the same idea?

28. Which route feels more natural if you already know Laplace transforms well: differentiating $F(a)$ first, or rewriting $F(a)$ as a Laplace transform of $\frac{\sin x}{x}$? Explain your answer.

29. Why is the value $\pi/2$ plausible from the formula

$$F(a) = \frac{\pi}{2} - \arctan a$$

when $a \rightarrow 0^+$?

30. Explain in words why oscillation of $\sin x$ is essential for convergence on $(1, \infty)$.

31. Sketch the graph of $\frac{\sin x}{x}$ and explain how the alternating positive and negative lobes fit the value of the integral.

32. Compute

$$\int_0^\infty \frac{\sin(bx)}{x} dx$$

for $b > 0$, and verify your formula by checking the special case $b = 1$.

8 Integrals and Euler's Method

1. Consider the initial value problem

$$y'(x) = f(x, y(x)), \quad y(a) = y_0.$$

Write down the forward Euler update on a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

and explain why one step has the form

$$y_{i+1} - y_i \approx f(x_i, y_i) \Delta x_i.$$

2. Why is a single Euler update naturally interpreted as an accumulation of a local rate over a small interval?
3. Explain why it is reasonable to call

$$y_{i+1} = y_i + f(x_i, y_i) \Delta x_i$$

an integration step rather than merely an algebraic update.

4. Sum the Euler increments over many steps and derive

$$y_n = y_0 + \sum_{i=0}^{n-1} f(x_i, y_i) \Delta x_i.$$

What is the conceptual meaning of this formula, and how does the initial value y_0 enter as the starting accumulated amount?

5. Why does the exact solution of

$$y'(x) = f(x, y(x)), \quad y(a) = y_0,$$

satisfy the integral equation

$$y(x) = y_0 + \int_a^x f(t, y(t)) dt?$$

How does Euler's method mimic this identity?

6. Why is the solution curve of an ODE often called an integral curve? What is being integrated, and in what sense is the curve determined by that integration?
7. For a general first-order ODE

$$y' = f(x, y),$$

what information does the slope field encode? Why does prescribing an initial value

$$y(a) = y_0$$

select one particular curve from that field?

8. In the autonomous case

$$y' = F(y),$$

how does the interpretation of an integral curve differ from the case

$$y' = f(x, y)?$$

What remains the same?

9. Show that an ordinary definite integral is the special case of the ODE

$$y'(x) = g(x), \quad y(a) = y_0,$$

where the right-hand side does not depend on y . What is the exact solution?

10. Let G be one antiderivative of g , so that

$$G'(x) = g(x).$$

Why are all solutions of

$$y'(x) = g(x)$$

of the form

$$y(x) = G(x) + C?$$

How does the initial condition

$$y(a) = y_0$$

determine the value of C ?

11. Compare the slope field of a general ODE

$$y' = f(x, y)$$

with the slope field of

$$y' = g(x).$$

Why is the latter vertically uniform, meaning that at a fixed x the slope is the same at every height y ?

12. Why does the vertical uniformity of the slope field for

$$y' = g(x)$$

imply that all integral curves are vertical translates of one another? How does this connect to the additive constant C in

$$y = G(x) + C?$$

13. Apply Euler's method to

$$y'(x) = g(x), \quad y(a) = y_0.$$

Why does the scheme become

$$y_{i+1} = y_i + g(x_i) \Delta x_i?$$

14. Iterate the previous update and show that

$$y_n = y_0 + \sum_{i=0}^{n-1} g(x_i) \Delta x_i.$$

Why is this now literally a Riemann sum plus the initial value, rather than only an Euler approximation?

15. What special feature makes the discrete accumulation for

$$y' = g(x)$$

telescope exactly into

$$y_n - y_0 = \sum_{i=0}^{n-1} (y_{i+1} - y_i)?$$

Why does the same telescoping identity not by itself turn a general Euler method for

$$y' = f(x, y)$$

into a classical Riemann sum for a known function of x alone?

16. In the case

$$y'(x) = g(x),$$

under what conditions on the partition and sample points does

$$\sum_{i=0}^{n-1} g(x_i) \Delta x_i$$

converge to

$$\int_a^b g(x) dx?$$

State the connection with the definition of Riemann integrability.

17. If

$$y'(x) = f(x, y)$$

does depend on y , what breaks in the attempt to interpret Euler's method as an ordinary Riemann sum in the variable x alone? Which additional unknown quantity is being approximated at every step?

18. Compare these two viewpoints:

$$\int_a^b g(x) dx \quad \text{and} \quad y(b) - y(a) \quad \text{for} \quad y' = f(x, y).$$

In what sense are both global accumulations of local data, and in what sense is the second problem structurally more implicit?

19. If two solutions of

$$y' = g(x)$$

have different initial values, why does their difference remain constant in x ? Why is the analogous statement usually false for a general ODE

$$y' = f(x, y)?$$

9 The Connection Between the Net Change Theorem, Differential Equations, and Euler's Method

The connection between the net change theorem, differential equations, and Euler's method is structural. They are different views of the same fundamental operation:

- Differential equations describe *local rates of change*.
- Integration accumulates those local changes into a *global change*.
- Euler's method numerically performs that accumulation step-by-step.

The net change theorem is the bridge connecting these ideas.

Suppose that

$$F'(t) = f(t).$$

Then the net change theorem states:

$$F(b) - F(a) = \int_a^b f(t) dt \quad \text{or} \quad F(b) = F(a) + \int_a^b f(t) dt.$$

The interpretation is:

- $f(t)$ is the instantaneous rate of change,
- the integral accumulates infinitesimal changes,
- the result is the total change in F .

Now consider a differential equation:

$$y'(t) = f(t, y(t)).$$

This equation states that the instantaneous change of the unknown function y is determined by the function f .

Applying the net change theorem to the unknown solution y , we integrate both sides from t_0 to t :

$$y(t) - y(t_0) = \int_{t_0}^t f(s, y(s)) ds.$$

If the initial value is

$$y(t_0) = y_0,$$

then:

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds.$$

This equation has a direct interpretation:

- Start with the initial value y_0 ,
- accumulate all infinitesimal changes $f(s, y(s)) ds$,
- obtain the final value $y(t)$.

Thus, a differential equation is fundamentally an accumulation law.

9.1 Euler's Method as Numerical Integration

Euler's method arises by approximating the integral with a Riemann sum. Take a small step size h . Over the interval $[t_n, t_n + h]$,

$$y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_n+h} f(s, y(s)) ds.$$

Euler's approximation assumes that the slope remains approximately constant over this short interval:

$$f(s, y(s)) \approx f(t_n, y_n).$$

Therefore,

$$\int_{t_n}^{t_n+h} f(s, y(s)) ds \approx h f(t_n, y_n).$$

Substituting this approximation gives Euler's update rule:

$$y_{n+1} = y_n + h f(t_n, y_n).$$

This is not merely inspired by integration. It *is* numerical integration. Each Euler step performs the following process:

1. Measure the current rate of change,
2. assume it remains roughly constant over a short interval,
3. accumulate the resulting small change,
4. repeat.

Thus, every Euler step is a discrete version of the net change theorem:

$$\text{new value} = \text{old value} + \text{accumulated change}.$$

This explains why:

- numerical ODE solving is often called *integration*,
- solution curves are called *integral curves*,
- ODE solvers are often called *integrators*.